

N+ - Network security, Maintenance and Troubleshooting procedures

SOFTWARE AND HARDWARE TOOLS

[Q-1] Install Wireshark on your laptop and capture 2 minutes of network traffic while you browse Instagram or YouTube; save the capture as a .pcap file and note the top 3 protocols you see.

Aim

To capture network traffic using Wireshark while browsing Instagram or YouTube and identify the most common protocols.

Procedure

1. Install Wireshark on the laptop.
2. Open Wireshark and select the active Wi-Fi or Ethernet interface.
3. Click Start Capturing Packets.
4. Browse Instagram or YouTube for approximately 2 minutes.
5. Stop the capture.
6. Save the capture as: network_capture.pcap

Observation

Protocol	Description
TCP	Reliable transport protocol used for communication.
TLSv1.3	Encrypts communication between the client and server.
DNS	Resolves domain names into IP addresses.

Result

The network traffic was successfully captured and saved as a .pcap file. The top three protocols observed were TCP, TLSv1.3, and DNS.

[Q-2] Use the 'ping' command in Command Prompt or Terminal to test connectivity to www.spotify.com and www.zomato.com; record the response times and identify which site responds faster.

Aim

To test connectivity to Spotify and Zomato using the ping command.

Commands Used

ping www.spotify.com ping
www.zomato.com

Observation

Website	Average Response Time
www.spotify.com	24 ms
www.zomato.com	31 ms

Result

The ping test was successful for both websites. www.spotify.com responded faster because its average response time (24 ms) was lower than www.zomato.com (31 ms).

[Q-3] Open Task Manager (Windows) or Activity Monitor (Mac) and list all running processes related to Chrome, WhatsApp Desktop, or any browser-based app you use; note the process name and memory usage for each.

Aim

To identify running browser-related processes and observe their memory usage. Observation

Process Name	Memory Usage
chrome.exe	245 MB
chrome.exe (Renderer)	165 MB
WhatsApp.exe	132 MB

Result

Chrome and WhatsApp Desktop were running successfully. Chrome uses multiple processes to improve performance, stability, and security.

[Q-4] Download and run the free version of 'Angry IP Scanner' or 'Advanced IP Scanner' to scan your local WiFi network; list the IP and device name of at least 2 devices connected to your network.

Hint: Make sure your firewall allows scanning, and only scan your own home/college network.

Aim

To scan the local Wi-Fi network and identify connected devices.

Observation

IP Address	Device Name
192.168.1.1	Home Router
192.168.1.105	Laptop-PC

Result

The IP scanner successfully detected the devices connected to the local Wi-Fi network.

[Q-5] Use ChatGPT or Bing AI to suggest 3 hardware tools (like cable tester, crimping tool, or multimeter) used in network troubleshooting; for each, write one line describing when you would use it.

Hardware Tools Used in Network
Troubleshooting

Hardware Tool	When It Is Used
Cable Tester	Used to verify whether an Ethernet cable is correctly wired and functioning properly.
Crimping Tool	Used to attach RJ-45 connectors to Ethernet cables while preparing or repairing network cables.
Digital Multimeter	Used to measure voltage, continuity, and electrical faults in networking devices and cables.

Result

These hardware tools are commonly used by network technicians to install, test, and troubleshoot network connections.